Ashwin Balaji Arun Kumar

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Location: Dortmund, Germany

Education

Fachhochschule Dortmund – University of Applied Sciences and Arts Dortmund, DE
Master of Engineering (M.Eng.) in Embedded Systems Engineering 10/2025 – Current
(Results pending for first semester)

Anna University Coimbatore, India
Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering 08/2018 – 06/2022
GPA: 1.5 / 4.0 (German equivalent)

Skills

Languages & Tools: C, C++, Python, Assembly (Fundamentals), Matlab/Simulink, Git, SVN
Hardware: TI C2000 (TMS320F), Renesas RL78, STM32 (F4 Series), ESP32, RP2040
Communication Protocols: I2C, UART, SPI, Modbus, RS485, MQTT
Build Systems: Latex, Linker Scripts, Make, CMake, Makefiles
Quality & Standards: Static Analysis (Coverity), Failure Mode and Effects Analysis (FMEA), IEC standards

Work Experience

Schneider Electric Mumbai, India
Deputy Manager – Firmware 08/2023 – 09/2025

- Spearheaded **3-20 HP VFD** firmware development in C, managing the full lifecycle on TI and RL78 MCUs.
- **Architected** a modular application layer, decoupling control logic from hardware using **HAL** for portability.
- Developed a high-efficiency (98%) **P&O MPPT** algorithm for DC solar pumping systems.
- Optimized **short-circuit protection** to **200ns** using Trip Zones, thereby **reducing field failures.**
- Integrated **static analysis** into a CI/CD pipeline, ensuring **10+** high-quality firmware releases.
- Replaced physical EEPROM with **flash-based EEPROM emulation**, significantly reducing unit **BOM cost**.

Graduate Engineer Trainee – Electromechanical 07/2022 – 08/2023

- Supported **6A – 100A RCBO** range lifecycle by managing Design changes and customer complaints.
- Reduced short-circuit **let through energy by 4%** through contact design optimization.(as per IEC 61009-1)
- Identified 30+ failure cases in DFMEA and **resolved 2 failures** with RPN>144.

Projects

i-Practace – Automatic Tennis ball throwing machine

- Secured funding of **€9,900** (Rs.9,48 lakhs) grant from Department of Science & Technology, Govt of India
- Managed the overall POC Development, including hardware selection, firmware including AC/DC motor control, sensor integration, and IoT development.

Patents and Publications

- **Automated Tennis game coaching device** - Indian Design Patent Number - 367774-001 (Journal No-20/2023)
- Comparative Analysis on Industrial IoT Communication Protocols and its Future Directives - IEEE XPLORE

Languages

German: A2 (Actively pursuing B1) **English:** C1